

Thomas ALLARD

PhD in Condensed Matter Theory

Research interests: time-varying media, light-matter interaction, topological photonics

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29 years old
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RESEARCH EXPERIENCES

- 2024/10 -** UNIVERSIDAD AUTÓNOMA DE MADRID.
Postdoctoral researcher in the group of P. A. Huidobro (IFIMAC). Researches in the framework of the ERC TIMELIGHT (Time-Varying Nanophotonics for New Regimes of Light-Matter Interactions).
- 2023/10 -** UNIVERSITÉ DE STRASBOURG.
2024/10 **Postdoctoral position (temporary lecturer)** in the group of G. Weick (IPCMS), in collaboration with D. Hagenmüller (CESQ).
- 2020/10 -** UNIVERSITÉ DE STRASBOURG.
2023/10 **PhD in Condensed Matter Theory**, supervised by G. Weick (IPCMS): *Disorder and topology in strongly-coupled light-matter systems*.
- 2020/04 -** UNIVERSITÉ DE STRASBOURG.
2020/08 **Master thesis**, supervised by G. Weick (IPCMS): *Quantum theory of polaritons in nanoplasmonics*.

EDUCATION

- 2024** UNIVERSITÉ DE STRASBOURG.
Qualification to “Maitre de conférence”, section 28.
- 2019-2020** ECOLE NORMALE SUPÉRIEURE DE PARIS.
Master degree in Physics, 2nd year: International Center for Fundamental Physics, Theoretical Physics track (with honours).
- 2018-2019** UNIVERSITÉ DE STRASBOURG.
Higher education competitive teaching examination: “Préparation à l’Agrégation”
➤ Admitted to the “Agrégation de Sciences Physiques option Physique”.
- 2017-2018** UNIVERSITÉ DE STRASBOURG.
Master degree in Physics, 1st year (with honours)
➤ Exchange semester with LUDWIG MAXIMILIANS UNIVERSITÄT, MÜNCHEN.
- 2016-2020** UNIVERSITÉ DE STRASBOURG.
“Magistère de Physique Fondamentale” (with honours).
- 2014-2017** UNIVERSITÉ DE STRASBOURG.
Bachelor degree in Physics (with honours).

TEACHING AND MENTORING EXPERIENCES

- 2024-** UNIVERSIDAD AUTÓNOMA DE MADRID.
Participation to the supervision of the PhDs of J. E. Sustaeta-Osuna and A. Caballero.
- 2020-2024** UNIVERSITÉ DE STRASBOURG.
280 teaching hours as:
- Lecturer for the course Mathematics for Physicists (1st year).
- Oral examinations and exam creations in Physics (1st year).
- Teaching assistant in Laboratory Works (1st year), Computer Science (1st year), Advanced Classical Mechanics (2nd year) and Statistical Physics (3rd and 4th year).

PUBLICATIONS AND PREPRINTS

- [1] **T. F. Allard** and G. Weick, Quantum theory of plasmon polaritons in chains of metallic nanoparticles: From near- to far-field coupling regime, *Phys. Rev. B* **104**, 125434 (2021).
- [2] **T. F. Allard** and G. Weick, Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons, *Phys. Rev. B* **106**, 245424 (2022).
- [3] **T. F. Allard** and G. Weick, Multiple polaritonic edge states in a Su-Schrieffer-Heeger chain strongly coupled to a multimode cavity, *Phys. Rev. B* **108**, 245417 (2023).
- [4] **T. F. Allard** and G. Weick, Mirror-induced effects in cavity polaritonics: Influence on edge states, *Phys. Rev. B* **110**, 125423 (2024).
- [5] A. Miguel-Torcal, **T. F. Allard**, P. A. Huidobro, F. J. García-Vidal and A. I. Fernández-Domínguez, Constructing Qubit Edge States by Inverse-Designing the Electromagnetic Environment, *ACS Photonics* **12**, 10, 5434-5442 (2025).
- [6] **T. F. Allard**, J. E. Sustaeta-Osuna, F. J. García-Vidal and P. A. Huidobro, Broadband Dipole Absorption in Dispersive Photonic Time Crystals *Phys. Rev. Lett.*, In Press (2026).
- [7] J. E. Sustaeta-Osuna, **T. F. Allard**, F. J. García-Vidal and P. A. Huidobro, Near-Field Gain and Far-Field Control via a Plasmonic Time Crystal Slab *Phys. Rev. Lett.*, In Press (2026).
- [8] **T. F. Allard** and G. Weick, Reentrant localization transition in a dimerized quasiperiodic dipolar chain, [arXiv:2501.16514](#).
- [9] A. Caballero, **T. F. Allard** and P. A. Huidobro, Interface states in space-time photonic crystals: topological origin, propagation and amplification [arXiv:2510.18523](#).
- [10] F. Maurer, **T. F. Allard**, Y. Todorov, G. Weick, D. Hagenmüller, Microscopic quantum description of surface plasmon polaritons: revealing intrinsic ultrastrong light-matter coupling [arXiv:2601.11297](#).

PEER REVIEWING ACTIVITY

- Referee for Phys. Rev. A, Phys. Rev. B, Phys. Rev. Lett., Nanophotonics, Nature.

PRESENTATIONS

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| 2026/04 | IPCMS QUANTUM SCIENCES SEMINAR, STRASBOURG, FRANCE.
Invited talk: <i>Photonics of time-varying media.</i> |
| 2026/04 | SPIE PHOTONICS EUROPE, STRASBOURG, FRANCE.
Contributed talk: <i>Broadband dipole absorption in dispersive photonic time crystals.</i> |
| 2025/10 | PLENARY SESSION GDR ONDES, BESANÇON, FRANCE.
Poster: <i>Broadband dipole absorption in dispersive photonic time crystals.</i> |
| 2025/10 | MOLECULAR POLARITONICS, MIRAFLORES, SPAIN.
Invited talk: <i>Topology in the strong coupling regime: polaritonic and non-polaritonic effects.</i> |
| 2025/09 | METAMATERIALS, AMSTERDAM, NETHERLANDS.
Contributed talk: <i>Broadband dipole absorption in dispersive photonic time crystals.</i> |
| 2025/06 | STRONG COUPLING IN ORGANIC MOLECULES, ODENSE, DENMARK.
Poster: <i>Mirror-induced effects in cavity polaritonics: Influence on edge states.</i> |
| 2025/06 | SPANISH CONFERENCE ON NANOPHOTONICS, MADRID, SPAIN.
Poster: <i>broadband dipole absorption in dispersive photonic time crystals.</i> |
| 2025/05 | WAVES IN TIME-VARYING MEDIA WORKSHOP, MADRID, SPAIN.
Poster: <i>Broadband dipole absorption in dispersive photonic time crystals.</i> |
| 2025/01 | GEFES, OVIEDO, SPAIN.
Poster: <i>Mirror-induced effects in cavity polaritonics: Influence on edge states.</i> |
| 2025/01 | QNANOLIGHT WORKSHOP, MIRAFLORES, SPAIN.
Invited talk: <i>Photonics of time-varying media.</i> |
| 2024/10 | IFIMAC INFORMAL SEMINAR, MADRID, SPAIN.
Invited talk: <i>Topology in the strong-coupling regime.</i> |
| 2024/05 | SPRING MEETING OF THE EMRS, STRASBOURG, FRANCE.
Contributed talk: <i>Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons.</i> |

- 2023/11** PLENARY SESSION GDR MESOSCOPIC QUANTUM PHYSICS, AUSSOIS, FRANCE.
Poster: *Multiple polaritonic edge states in a Su-Schrieffer-Heeger chain strongly coupled to a multimode cavity.*
- 2023/06** WORKSHOP ON TOPOLOGICAL PHOTONICS, MADRID, SPAIN.
Poster: *Multiple polaritonic edge states in a Su-Schrieffer-Heeger chain strongly coupled to a multimode cavity.*
- 2022/11** PLENARY SESSION GDR MESOSCOPIC QUANTUM PHYSICS, AUSSOIS, FRANCE.
Poster: *Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons.*
- 2022/09** WORKSHOP ON MOLECULAR POLARITONICS, STRAUBING, GERMANY.
Poster: *Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons.*
- 2022/06** WORKSHOP ON TOPOLOGICAL PHOTONICS, DONOSTIA-SAN SEBASTIÁN, SPAIN.
Contributed talk: *Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons.*
- 2022/04** EUROPEAN SPRING SCHOOL ON QUANTUM SCIENCE, STRASBOURG, FRANCE.
Contributed talk: *Disorder-enhanced transport in a chain of lossy dipoles strongly coupled to cavity photons.*
- 2021/11** PLENARY SESSION GDR MESOSCOPIC QUANTUM PHYSICS, AUSSOIS, FRANCE.
Contributed talk: *Absence of Anderson localization in 1D due to cavity photons.*
- 2021/10** SUMMER SCHOOL ON MESOSCOPIC QUANTUM PHYSICS, CARGÈSE, FRANCE.
Poster: *Absence of Anderson localization in 1D due to cavity photons.*
- 2021/07** PARIS-SACLAY/MUNICH SUMMER SCHOOL ON SURFACE PLASMONS, SACLAY, FRANCE.
Poster: *Quantum theory of plasmon-polaritons in chains of metallic nanoparticles.*
- 2020/11** PLENARY SESSION GDR MESOSCOPIC QUANTUM PHYSICS, AUSSOIS, FRANCE.
Poster: *Polaritons in periodic chains of metallic nanoparticles: a QED approach.*

LANGUAGES

French: Native.
 English: Fluent.
 German: Intermediate.
 Spanish: Intermediate.
 Programming: $\text{\LaTeX}$, UNIX, Python.

HOBBIES

In- and outdoor climbing, hiking, volunteer activities in a cultural association.